

AutoLISP / Visual LISP Spec Reference (alref)

User Manual

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1 Introduction

`alref` is the spec-reference library shipped with `autolisp-spec`. It has two parallel surfaces:

- `alref.el` — an Emacs minor-mode + commands (`M-x alref-lookup`, `alref-autodoc-mode`).
- `alref.lsp` — an AutoLISP library you can load into any conforming runtime (AutoCAD, BricsCAD, clautolisp) for interactive use at the AutoLISP REPL.

Both consume the paged spec artefacts produced by `scripts/build-paged-spec.el` (`build/pages/`, `build/html/`, `build/info/`) and expose the same vocabulary: look up a symbol to find its description, search the index by substring, walk chapter intros, etc.

2 Installation

The paged spec data, the Emacs lib, and the AutoLISP lib install under:

`$(PREFIX)/share/autolisp-spec/pages/` data files (alref reads) `$(PREFIX)/share/doc/autolisp-spec/` reference org + pdf docs `$(PREFIX)/share/info/` `autolisp-spec.info`
`$(PREFIX)/share/emacs/site-lisp/autolisp-spec/` `alref.el` `$(PREFIX)/share/autolisp/site-lisp/autolisp-spec/` `alref.lsp`

Run `sudo make install` from this subproject to lay everything down.
The default PREFIX is `/opt/local`; override on the command line.

3 alref.el — Emacs

After installation, add the site-lisp directory to your `load-path` (distros usually do this automatically) and:

(require 'alref)

Then:

- `M-x alref-lookup RET name RET` — completing read over every documented symbol, opens the matching HTML page in your browser.
- `M-x alref-autodoc-mode` — buffer-local minor mode that shows the operator signature of the symbol at point in the minibuffer, like `slime-autodoc`. Picks up the `[A]` / `[B]` / `[C]` / `[ABC]` compatibility tag from the symbol's Compatibility section when the page has one.
- Customise the data root via `M-x customize-variable RET alref-spec-directory` or set it in your init file.

4 alref.lsp — AutoLISP

At the AutoLISP REPL, load the library (typically from a user init file):

(load "opt/local/share/autolisp/site-lisp/autolisp-spec/alref.lsp") (alref-set-root "opt/local/share/autolisp-spec")

Then call:

- (`alref-lookup "pattern"`) — print every documented symbol whose name contains PATTERN (case-insensitive).
- (`alref-apropos-list "pattern"`) — same, but returns the list of names rather than printing.
- (`alref-apropos "pattern"`) — print one symbol per line with its live state and a five-column [ABC] coverage indicator:
NAME<TAB>Function<TAB>[ABC] NAME<TAB>Operator<TAB>[ABC]
NAME<TAB>Variable<TAB>[ABC]<TAB>VALUE NAME<TAB>Variable<TAB>[ABC]<TAB>
The indicator's letters mean: A=spec'd by AutoCAD, B=spec'd by BricsCAD, C=implemented in this image (the running runtime).
- (`alref-describe NAME`) — print the spec page for NAME (a symbol, a string name, a chapter number, or a chapter title). When NAME isn't in the spec but is live in the image, prints the single-line apropos display instead.
- (`alref-documentation NAME`) — return the spec page text as a string (instead of printing). Returns "not documented" for image-only symbols and nil for names neither in the spec nor in the image.
- (`alref-help "string"`) — search the body of every documented page for STRING and return the list of matching page basenames.

5 The [ABC] coverage indicator

The indicator surfaces in two places:

- `alref.el`'s `alref-autodoc-mode` shows it at the end of the minibuffer signature line.
- `alref.lsp`'s `alref-apropos` shows it as a fixed-width five-character column.

[A] spec'd by AutoCAD only [B] spec'd by BricsCAD only [C] implemented in this image only (no spec entry) [AB] spec'd by both vendors, not in this image [AC] spec'd by AutoCAD, implemented in this image [BC] spec'd by BricsCAD, implemented in this image [ABC] spec'd by both vendors AND in this image

The data comes from the per-spec-entry **Availability** section (parsed at paged-build time into the 4th column of `symbols.txt`) plus the runtime's `atoms-family` (for the C bit).

6 Re-building the data

After editing the canonical spec (`autolisp-visual-lisp-specification-draft.org`), regenerate the paged artefacts:

```
make pages ## emits build/pages/*.txt + symbols.txt + index.txt
make html ## emits build/html/*.html
make info ## emits build/info/autolisp-spec.info
make documentation ## also rebuilds the PDF
make install ships every artefact.
```

7 See also

- `autolisp-visual-lisp-specification-draft.org` — the canonical source of truth for the spec content.
- `build/pages/` — plain-text per-section pages alref reads.
- `scripts/build-paged-spec.el` — the splitter.
- `issues/closed/apropos-non-alref-symbols.issue` — origin of the image-symbol union + [ABC] indicator.
- `issues/closed/autolisp-spec-paged-version.issue` — origin of the paged-build infrastructure.

8 Synchronisation

Per `AGENTS.md`, changes to the spec source / paged build / alref libs / this manual must propagate.